Projector-Resolved Seed Calculus III

Typed Edge Operators, Connection Defects, and Obstruction Packets

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Abstract

This paper separates three structures that are easy to conflate. First, a finite packet graph supports typed elementary operators: an edge from packet i to packet j contributes only to the (j, i) matrix block. Noncomposable products therefore vanish, and powers of the total transfer operator expand into ordered operator words whose surviving terms are directed walks. Lean checks the recursive, explicit-word, and walk-filtered forms, the endpoint coefficient formula, and fixed-length and orbit-support reachability obstructions. Second, exact packet coordinates on one ambient module carry a canonical transport

$$J_{b\leftarrow a} = \mathcal{A}_b \mathcal{S}_a.$$

Its frame-level identity, cocycle, round-trip, and triangle-loop laws are checked in Paper III; the released formalization also checks composition and naturality for arbitrary functions between exact coordinate systems. Linear packet-coordinate counterparts remain available from the frozen contract of Paper II. Consequently canonical coordinate transport has no holonomy. Third, a connection $\Gamma_{b\leftarrow a}$ is additional, noncanonical transport data. Lean checks both a minimal self-normalized API and a heterogeneous reversible API, including the additive defect–holonomy identity, finite-path collapse under flatness, gauge covariance, and gauge equivalence of flat pair connections to canonical transport. It also checks exact finite block reconstruction of triangle obstructions, the four equivalent vanishing criteria, packet-supported witnesses, and packet-preserving block-gauge covariance. None of these results identifies Γ with J by definition. Finally, Lean supplies finite unary–binary operation-tree syntax, arbitrary semantics, locality, and candidate-specific obstruction certificates. Colored trees, multilinear packet expansions, and nonlinear Faà di Bruno defects remain roadmap items. No general nonlinear chain rule, analytic convergence theorem, or physical interpretation is claimed.

Final verification boundary. Version 1.0.1 is theorem-complete for its explicitly scoped claims. All thirty theorem-like environments are formally backed: twenty-six by compiled Paper III declarations and four by the frozen Papers I–II contracts. The Paper III scope is checked with Lean 4 v4.34.0-rc2 against Mathlib revision `aeb72f1e9981b6d35b426b16dca7759be3e7ba06` [3, 4] from `EdgeOperators.lean`, `EdgeExtensions.lean`, `Transport.lean`, `ReversibleConnections.lean`, `CoordinateNaturality.lean`, `HolonomyBlocks.lean`, and `NonlinearObstructions.lean`; the declaration-level ledger is given in section 12. Roadmap items are explicit exclusions, not completion claims. Definitions do not count as theorems. In particular, the words *connection*, *curvature*, and *holonomy* refer only to the algebraic APIs defined here. Formal verification establishes conformance to the stated specifications; it does not by itself establish novelty or priority.

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1 Purpose and status discipline

Paper I fixed typed seeds, operator grammars, exact projector packets, generated hulls, and reconstruction [1]. Paper II added coherent refinement coordinates, weighted transfer, arithmetic separation, and linear defect identities [2]. The present paper begins the next layer: explicit edge words, genuinely optional connection data, and packet-local obstructions.

The central distinction is categorical rather than terminological. Canonical coordinate transport compares two exact descriptions of the *same* ambient object and is forced by analysis and synthesis. A connection chooses transports that may carry extra modeling, discretization, or gauge information. The former obeys a cocycle theorem; failure of the latter to obey a cocycle is meaningful data. Treating both maps as one symbol would make holonomy either tautologically trivial or logically unfounded.

1.1 Statement labels

Every substantial theorem, lemma, corollary, proposition, definition/API, and roadmap item carries status evidence drawn from one or more of three labels. A statement whose separate clauses use both a Paper III result and an inherited bridge carries both applicable labels. Illustrative examples are exempt.

S1. Lean-backed in Paper III: the displayed scope is no stronger than the named compiled declaration in the released Paper III module.

S2. Inherited Lean-backed: the displayed result is supplied by the frozen formal contract of Paper I or II.

S3. Roadmap only: the text states a future interface or theorem target and is excluded from the completion denominator.

Boundary 1.1 (Final theorem denominator). The completion denominator is the thirty theorem-like environments in the paper: twenty-six resolve to compiled Paper III declarations and four resolve to the frozen Papers I–II contracts. The authoritative theorem map names the exact declarations used by each statement. The kernel audit checks those targets and supporting theorems, while a fail-closed gate scans the complete compiled `Komorebi.PaperIII` namespace. Roadmap items are excluded from the denominator and remain visibly marked.

2 Finite typed packet graphs

2.1 Orientation and matrix convention

Let V be a finite packet type, E a finite edge type, and R a semiring. A directed multigraph is a pair of endpoint functions

$$\mathrm{src}, \mathrm{tgt} : E \longrightarrow V.$$

Parallel edges remain distinct elements of E. Rows are targets and columns are sources. Thus an edge $e : i \to j$ is represented by a matrix supported at row j, column i.

Definition 2.1 (Packet graph and scalar edge system)**.** For $a, b \in V$ and $r \in R$, define

$$\mathrm{packetMatrix}(a, b; r)_{yx} = \begin{cases} r, & y = b \text{ and } x = a, \\ 0, & \text{otherwise.} \end{cases}$$

An edge system is a coefficient function $c : E \to R$. Its elementary edge operator and total transfer operator are

$$L_e = \mathrm{packetMatrix}(\mathrm{src}(e), \mathrm{tgt}(e); c(e)), \qquad T = \sum_{e \in E} L_e.$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph;Komorebi.PaperIII.EdgeOperators.PacketGraph.packetMatrix;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp`

The convention is execution ordered. In $L_f L_e$, the rightmost operator L_e acts first. For a written word $(e_1, \ldots, e_n)$, e_1 acts first and the operator product is $L_{e_n} \cdots L_{e_1}$.

Lemma 2.2 (Declared support)**.** *For every edge e, $(L_e)_{yx} = 0$ unless $y = \mathrm{tgt}(e)$ and $x = \mathrm{src}(e)$.*

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp_eq_zero_off_support`

Theorem 2.3 (Typed interface annihilation)**.** *If $\mathrm{tgt}(e) \neq \mathrm{src}(f)$, then*

$$L_f L_e = 0.$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp_mul_edgeOp_eq_zero`

Proof. For a matrix entry (y, x),

$$(L_f L_e)_{yx} = \sum_{z \in V} (L_f)_{yz} (L_e)_{zx}.$$

A potentially nonzero summand would require simultaneously $z = \mathrm{src}(f)$ and $z = \mathrm{tgt}(e)$, contrary to the hypothesis. □

Theorem 2.4 (Matched two-edge collapse)**.** *If $\mathrm{tgt}(e) = \mathrm{src}(f)$, then*

$$L_f L_e = \mathrm{packetMatrix}\big(\mathrm{src}(e), \mathrm{tgt}(f); c(f)c(e)\big).$$

Together with theorem 2.3, this gives the complete composable/noncomposable dichotomy for two edge operators.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp_mul_edgeOp_of_composable`

Proof. The matrix-product sum has exactly one potentially nonzero intermediate index, namely $\mathrm{tgt}(e) = \mathrm{src}(f)$. At that index the coefficients occur in operator order as $c(f)c(e)$, and the outer source and target are $\mathrm{src}(e)$ and $\mathrm{tgt}(f)$. □

Boundary 2.5 (Current fiber scope). The compiled Paper III model above has one scalar coordinate per packet. For a family of modules $(X_i)_{i \in V}$, an intrinsic edge map would instead have type $\ell_e : X_{\mathrm{src}(e)} \to X_{\mathrm{tgt}(e)}$, and its totalization on $\bigoplus_i X_i$ would be $\iota_{\mathrm{tgt}(e)} \ell_e \pi_{\mathrm{src}(e)}$. The same support proof applies, but this dependent fiber-valued generalization is not attributed to the released scalar Paper III edge-system modules.

Figure 1 summarizes the support convention and the composable/noncomposable word dichotomy used throughout the next section.

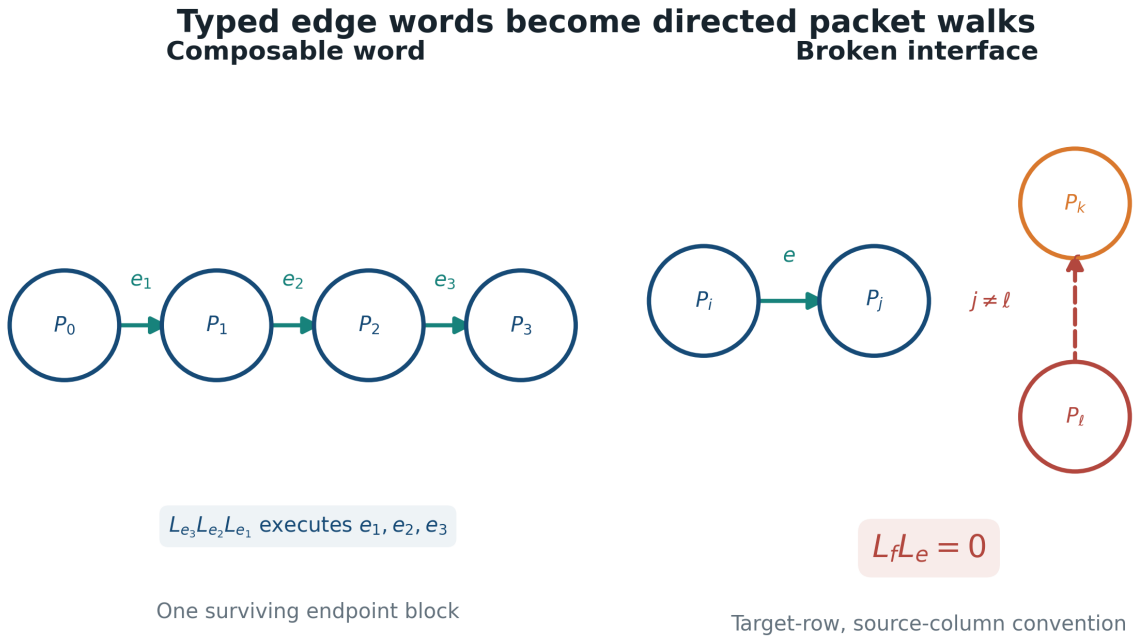


Figure 1: The target-row, source-column convention fixes execution order. Composable words collapse to endpoint blocks, while one broken packet interface annihilates the entire operator product.

3 Operator words and directed walks

3.1 Typed words

Definition 3.1 (Composable word and execution operator)**.** A finite word $w = (e_1, \dots, e_n)$ is composable when

$$\mathrm{tgt}(e_r) = \mathrm{src}(e_{r+1}) \qquad (1 \le r < n).$$

Set

$$L_w = L_{e_n} \cdots L_{e_1}, \qquad L_\varnothing = I.$$

This definition records the written order of execution even though matrix composition is read right to left.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.Composable;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp`

Theorem 3.2 (First broken interface)**.** *If a word begins* $(e, f, \dots)$ *and* $\mathrm{tgt}(e) \neq \mathrm{src}(f)$*, then its word operator is zero.*

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp_eq_zero_of_first_break`

Theorem 3.3 (Nonwalk words vanish)**.** *If* w *is not composable, then* $L_w = 0$*.*

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp_eq_zero_of_not_composable`

Proof. Choose the first broken adjacent pair. The corresponding consecutive factor is zero by theorem 2.3, and associativity absorbs that zero into the complete product. The Lean proof implements the same argument by induction on the list. □

3.2 The checked recursive expansion

Definition 3.4 (Recursive word expansion)**.** Define

$$\mathcal{W}_0 = I, \qquad \mathcal{W}_{n+1} = T\mathcal{W}_n.$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordExpansion`

Theorem 3.5 (Transfer power equals recursive word expansion)**.** *For every* $n \in \mathbb{N}$,

$$T^n = \mathcal{W}_n.$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_eq_wordExpansion`

Proof. The zero case is the identity matrix. If $T^n = \mathcal{W}_n$, then the left-successor power convention gives $T^{n+1} = TT^n = T\mathcal{W}_n = \mathcal{W}_{n+1}$. □

Remark 3.6. The formal module also defines an explicit sum over tuples $w : \mathrm{Fin}(n) \to E$ and a second sum in which noncomposable tuples are replaced by zero. It proves that the recursive, explicit, and walk-filtered matrices agree. These are separate declarations so that the finite enumeration claim is auditable rather than hidden in the name *word expansion*.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.explicitWordExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.walkExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordExpansion_eq_explicitWordExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.explicitWordExpansion_eq_walkExpansion`

3.3 Exact word-to-walk expansion

For a nonempty composable word $w = (e_1, \ldots, e_n)$, define its reverse coefficient product

$$c^{\leftarrow}(w) = c(e_n)c(e_{n-1}) \cdots c(e_1).$$

The reversal is essential over a noncommutative semiring.

Lemma 3.7 (One word is one endpoint block)**.** *If* $w = (e_1, \ldots, e_n)$ *is nonempty and composable, then*

$$L_w = \mathrm{packetMatrix}\big(\mathrm{src}(e_1), \mathrm{tgt}(e_n); c^{\leftarrow}(w)\big).$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.reverseCoefficient;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.terminalTarget;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp_eq_packetMatrix_of_composable`

Proof. Induct on n. The case $n = 1$ is the definition of L_e. Suppose the formula holds for the prefix $(e_1, \dots, e_n)$ and append f. Composability gives $\mathrm{tgt}(e_n) = \mathrm{src}(f)$. Multiplying the two packet matrices leaves one intermediate index and changes the coefficient to $c(f)c(e_n)\cdots c(e_1)$, with source $\mathrm{src}(e_1)$ and target $\mathrm{tgt}(f)$. □

Write $\mathrm{Walk}_n(i, j)$ for the finite set of length-n composable edge tuples whose first source is i and whose final target is j. For $n = 0$, it has one empty walk when $i = j$ and is empty otherwise; assign the empty walk the coefficient 1.

Theorem 3.8 (Finite operator word-to-walk expansion)**.** *For every $n \in \mathbb{N}$,*

$$T^n = \sum_{w \in E^n} L_w = \sum_{w \in E^n} \begin{cases} L_w, & w \text{ composable}, \\ 0, & w \text{ not composable}. \end{cases}$$

Parallel edges occur as distinct summands.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_eq_explicitWordExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_eq_walkExpansion`

Proof. Induction and finite distributivity give

$$\left(\sum_{e \in E} L_e\right)^n = \sum_{(e_1, \dots, e_n) \in E^n} L_{e_n} \cdots L_{e_1}.$$

Every noncomposable summand is zero by theorem 3.3, so inserting the composability filter does not change the sum. In Lean, tuple concatenation and the successor sum are handled by `tupleWordOp_snoc` and `explicitWordExpansion_succ`; the two bridge equalities then yield the displayed theorem. □

Corollary 3.9 (Endpoint coefficient walk sum)**.** *For all $n \in \mathbb{N}$ and $i, j \in V$,*

$$(T^n)_{ji} = \sum_{w \in \mathrm{Walk}_n(i,j)} c^{\leftarrow}(w).$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.Connects;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.endpointWalkCoefficientSum;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_apply_eq_sum_walks`

Proof. Take the (j, i) entry in theorem 3.8. By lemma 3.7, a surviving word contributes $c^{\leftarrow}(w)$ exactly when its endpoints are i and j, and contributes zero otherwise. At $n = 0$, the identity matrix gives the unique empty diagonal walk. □

Corollary 3.10 (Fixed-length reachability obstruction)**.** *If $\mathrm{Walk}_n(i, j) = \varnothing$, then $(T^n)_{ji} = 0$.*

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_apply_eq_zero_of_no_walk`

Proof. This is immediate from the empty sum in the preceding corollary. □

Corollary 3.11 (Orbit support obstruction)**.** *Let $S \subseteq V$, and let $x \in R^V$ be supported on S. If j is not reachable by a directed path from any vertex of S, then*

$$(T^n x)_j = 0 \qquad \text{for every } n \in \mathbb{N}.$$

Hence the T-orbit of x cannot generate a vector with nonzero j-coordinate.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.Reachable;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_mulVec_apply_eq_zero_of_unreachable_support`

Formal status: inherited Lean-backed result.
`Komorebi.PacketNaturalityCompletion.Reachability.algebraicOrbitHull_le_reachable_support;Komorebi.PacketNaturalityCompletion.Reachability.unreachable_nonzero_packet_obstruction`

Proof. Expand $(T^n x)_j = \sum_i (T^n)_{ji} x_i$. Terms with $i \notin S$ vanish by the support of x, and terms with $i \in S$ vanish by corollary 3.10. The final claim follows because the j-coordinate also vanishes on every finite linear combination of orbit vectors. □

4 Canonical exact coordinate transport

4.1 The Paper III frame model

Let every chart a carry a linear frame equivalence $F_a : M \xrightarrow{\sim} M$. The formal Paper III module defines transport from a to b by

$$J_{b \leftarrow a} = F_b^{-1} F_a.$$

This is the abstract frame form of *synthesize in chart a, then analyze in chart b.*

Theorem 4.1 (Exact frame transport is coherent)**.** *For every a, b, c,*

$$J_{a \leftarrow a} = \mathrm{id}, \qquad J_{a \leftarrow b} J_{b \leftarrow a} = \mathrm{id}, \qquad J_{c \leftarrow b} J_{b \leftarrow a} = J_{c \leftarrow a},$$

and therefore

$$J_{a \leftarrow c} J_{c \leftarrow b} J_{b \leftarrow a} = \mathrm{id}.$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.exactTransport_self;Komorebi.PaperIII.Transport.exactTransport_roundTrip;Komorebi.PaperIII.Transport.exactTransport_cocycle;Komorebi.PaperIII.Transport.exactTransport_triangleLoop`

Proof. Every identity is cancellation of adjacent $F_r F_r^{-1}$ factors. The Lean definitions use `LinearEquiv.trans`; the displayed formulas use the equivalent right-to-left function-composition notation fixed in this paper. □

Theorem 4.2 (Every finite canonical path telescopes)**.** *Let $p = (a_0, a_1, \ldots, a_n)$ be any finite vertex list and put*

$$J_p = J_{a_n \leftarrow a_{n-1}} \cdots J_{a_1 \leftarrow a_0}, \qquad J_{(a_0)} = \mathrm{id}.$$

Then

$$J_p = J_{a_n \leftarrow a_0}.$$

In particular, if $a_n = a_0$, then $J_p = \mathrm{id}$.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.pathEndpoint;Komorebi.PaperIII.Transport.exactPathTransport;Komorebi.PaperIII.Transport.exactPathTransport_eq_endpoint;Komorebi.PaperIII.Transport.exactPathTransport_loop`

Proof. Induct on the list of successive vertices and apply the cocycle in theorem 4.1 at each step. If the endpoint is the start, the endpoint transport is the identity. □

4.2 Exact packet coordinates

The graph model above resolves a fixed scalar matrix. Coordinate transport is a separate construction. Let X be an R-module, and let

$$\mathcal{P}^a = (P_i^a)_{i \in I_a}$$

be a finite complete family of pairwise orthogonal linear idempotents:

$$P_i^a P_j^a = \delta_{ij} P_i^a, \qquad \sum_{i \in I_a} P_i^a = I_X.$$

Define the exact packet coordinate module

$$\mathrm{Coord}(\mathcal{P}^a) = \{(x_i)_{i \in I_a} : P_i^a x_i = x_i\}.$$

Its analysis and synthesis maps are

$$\mathcal{A}_a(x) = (P_i^a x)_i, \qquad \mathcal{S}_a((x_i)_i) = \sum_i x_i.$$

Completeness and orthogonality give

$$\mathcal{S}_a \mathcal{A}_a = \mathrm{id}_X, \qquad \mathcal{A}_a \mathcal{S}_a = \mathrm{id}_{\mathrm{Coord}(\mathcal{P}^a)}.$$

Definition 4.3 (Canonical exact coordinate transport)**.** For two exact packet systems on the same ambient module, define

$$J^X_{b \leftarrow a} = \mathcal{A}_b \mathcal{S}_a : \mathrm{Coord}(\mathcal{P}^a) \longrightarrow \mathrm{Coord}(\mathcal{P}^b).$$

The arrow points from the source coordinate system a to the target coordinate system b.

Formal status: inherited Lean-backed result.
`Komorebi.PaperII.NestedRefinement.refineCoordinates;Komorebi.PaperII.NestedRefinement.coordinateRefinementEquiv`

Theorem 4.4 (Identity, inverse, and cocycle for canonical transport)**.** *For any exact packet systems a, b, c on X,*

$$J^X_{a \leftarrow a} = \mathrm{id}, \qquad \left(J^X_{b \leftarrow a}\right)^{-1} = J^X_{a \leftarrow b}, \qquad J^X_{c \leftarrow b} J^X_{b \leftarrow a} = J^X_{c \leftarrow a}.$$

Formal status: inherited Lean-backed result.
`Komorebi.PaperII.NestedRefinement.refineCoordinates_self;Komorebi.PaperII.NestedRefinement.coarsenCoordinates_refineCoordinates;Komorebi.PaperII.NestedRefinement.refineCoordinates_coarsenCoordinates;Komorebi.PaperII.NestedRefinement.refineCoordinates_comp;Komorebi.PaperII.NestedRefinement.coordinateRefinementEquiv_trans`

Proof. Insert $\mathcal{S}_b\mathcal{A}_b = \mathrm{id}_X$:

$$J^X_{c\leftarrow b}J^X_{b\leftarrow a} = \mathcal{A}_c\mathcal{S}_b\mathcal{A}_b\mathcal{S}_a = \mathcal{A}_c\mathcal{S}_a = J^X_{c\leftarrow a}.$$

The identity statement is $\mathcal{A}_a\mathcal{S}_a = \mathrm{id}$. Setting $c = a$ in the cocycle gives $J^X_{a\leftarrow b}J^X_{b\leftarrow a} = \mathrm{id}$; interchanging a and b gives the other composition, so the two transports are inverses. □

If $F : X \to Y$ is linear and $\mathcal{P}^a, \mathcal{P}^b$ are source packet systems while $\mathcal{Q}^r, \mathcal{Q}^s$ are target packet systems, put

$$\widehat{F}_{r\leftarrow a} = \mathcal{A}^Y_r F \mathcal{S}^X_a.$$

Theorem 4.5 (Canonical coordinate naturality)**.** *The exact coordinate representation preserves typed composition and satisfies*

$$J^Y_{s\leftarrow r}\widehat{F}_{r\leftarrow a} = \widehat{F}_{s\leftarrow b}J^X_{b\leftarrow a}.$$

Formal status: inherited Lean-backed result.
`Komorebi.PaperII.NestedRefinement.coordinateOperator_comp;Komorebi.PaperII.NestedRefinement.coordinateOperator_naturality`

Proof. Both sides reduce to $\mathcal{A}^Y_s F \mathcal{S}^X_a$ after inserting the appropriate synthesis–analysis identity. The composition law is the same calculation with the middle ambient module inserted. □

Corollary 4.6 (Canonical triangle holonomy is trivial)**.** *For every triple a, b, c,*

$$J^X_{a\leftarrow c}J^X_{c\leftarrow b}J^X_{b\leftarrow a} = \mathrm{id}.$$

Formal status: inherited Lean-backed result.
`Komorebi.PaperII.NestedRefinement.refineCoordinates_comp;Komorebi.PaperII.NestedRefinement.refineCoordinates_self`

Boundary 4.7 (Refinement is not reduction). Although Paper II uses the name *refineCoordinates*, J is an equivalence between two complete coordinate presentations. It neither truncates packets nor projects to a smaller state space. Any lossy restriction, interpolation, or learned identification is extra data and cannot inherit theorem 4.4 without proof.

5 A distinct noncanonical connection API

5.1 The minimal compiled API

The formal Paper III API starts with deliberately weak assumptions. All chart transports act on one module M, so chart labels distinguish routes but not Lean types.

Definition 5.1 (Self-normalized noncanonical connection)**.** A minimal connection is a family

$$\Gamma_{b\leftarrow a} \in \mathrm{Aut}_R(M) \qquad (a, b \in B)$$

with only

$$\Gamma_{a\leftarrow a} = \mathrm{id}_M.$$

Neither reverse-edge inversion nor the cocycle law is a field of the structure.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.Connection`

At a triple (a, b, c), the compiled cocycle predicate is

$$\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} = \Gamma_{c\leftarrow a}.$$

Flatness means this equality holds for every triple. The formal multiplicative relative defect and triangle holonomy are

$$D^{\times}_{c,b,a} = (\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a})\Gamma^{-1}_{c\leftarrow a}, \qquad H_{c,b,a} = \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a}.$$

The defect is based at the target c; the holonomy is based at a.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.Connection.CocycleAt;Komorebi.PaperIII.Transport.Connection.Flat;Komorebi.PaperIII.Transport.Connection.cocycleDefect;Komorebi.PaperIII.Transport.Connection.triangleHolonomy`

Theorem 5.2 (Checked flatness consequences)**.** *For the minimal connection API, flatness implies*

$$D^{\times}_{c,b,a} = \mathrm{id}_M, \qquad H_{c,b,a} = \mathrm{id}_M$$

for every a, b, c.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.Connection.cocycleDefect_eq_refl_of_flat;Komorebi.PaperIII.Transport.Connection.triangleHolonomy_eq_refl_of_flat`

Proof. Flatness replaces the two-edge route in $D^{\times}$ by the direct route, which cancels its inverse. For H, flatness first collapses $a \to b \to c$ to $a \to c$, then collapses $a \to c \to a$ to the normalized transport $a \to a$. □

Proposition 5.3 (The minimal API admits nontrivial holonomy)**.** *There is a connection on three points over $\mathbb{Q}$ for which transport $0 \to 1$ is multiplication by 2, every other unequal-endpoint transport is the identity, and*

$$H_{2,1,0}(1) = 2.$$

In particular $H_{2,1,0} \neq \mathrm{id}$.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.twistedThreePointConnection;Komorebi.PaperIII.Transport.twisted_triangleHolonomy_apply_one;Komorebi.PaperIII.Transport.twisted_triangleHolonomy_ne_refl`

Boundary 5.4 (Why the minimal defect and loop are distinct). Without reverse-edge inversion, $\Gamma_{a\leftarrow c}$ need not equal $\Gamma^{-1}_{c\leftarrow a}$. Therefore the multiplicative relative defect $D^{\times}_{c,b,a}$ and triangle holonomy $H_{c,b,a}$ are different comparisons. The formal module proves that global flatness trivializes both; it does not identify them for an arbitrary connection.

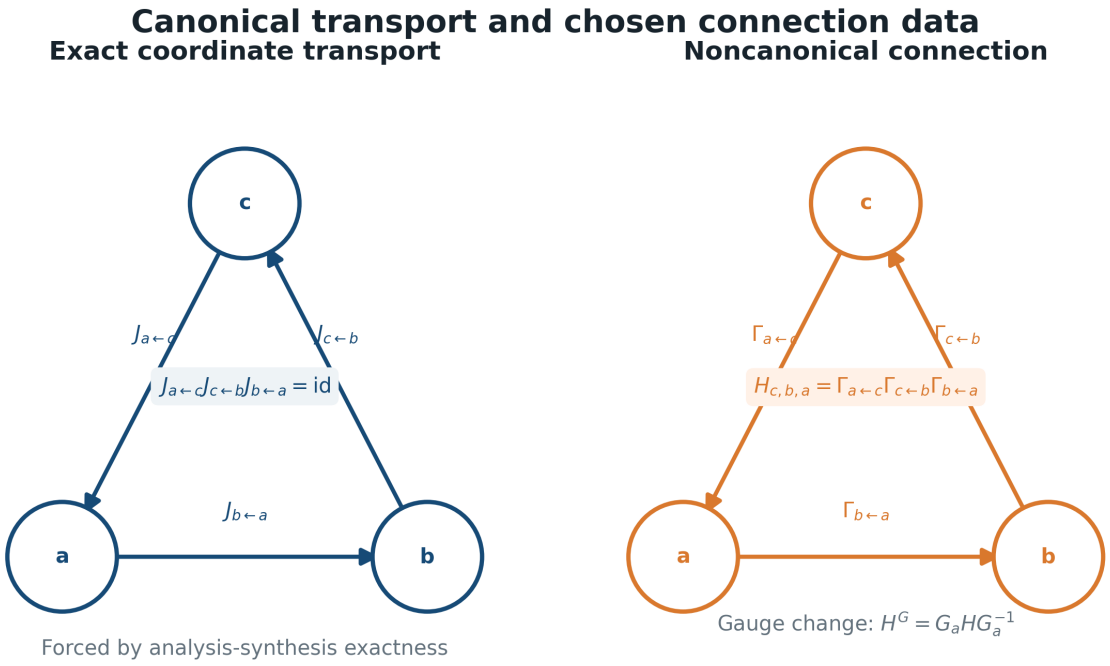


Figure 2: Exact coordinate transport and connection transport are distinct. The canonical loop telescopes to the identity. A chosen connection may carry nontrivial holonomy, which transforms by conjugation under gauge.

Figure 2 records this separation visually: the exact coordinate route telescopes, whereas the chosen connection route may carry a nonidentity loop.

5.2 A stronger reversible pair API

Let R be a commutative ring, let B be a type of charts or refinement states, and let each C_a be an R-module. The chart type is not assumed finite. The following definition is deliberately not derived from analysis and synthesis.

Definition 5.5 (Normalized reversible connection)**.** A normalized reversible connection is a family of linear equivalences

$$\Gamma_{b\leftarrow a} : C_a \xrightarrow{\sim} C_b \qquad (a, b \in B)$$

such that

$$\Gamma_{a\leftarrow a} = \mathrm{id}, \qquad \Gamma_{a\leftarrow b} = \Gamma_{b\leftarrow a}^{-1}.$$

No cocycle equation is included in the structure.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.ReversibleConnection`

This strengthens the homogeneous minimal connection with heterogeneous fibers and reverse-edge inversion. It remains additional data, not an alias for canonical coordinate transport.

Definition 5.6 (Calibration and flatness)**.** When the C_a are exact packet coordinates of one ambient module, the connection is *canonically calibrated* if

$$\Gamma_{b\leftarrow a} = J_{b\leftarrow a} \quad \text{for every } a, b.$$

It is *flat* if

$$\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} = \Gamma_{c\leftarrow a} \quad \text{for every } a, b, c.$$

The canonical connection is flat; a general flat connection is proved in theorem 6.3 to be gauge-equivalent to it.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.ReversibleConnection.CocycleAt;Komorebi.PaperIII.Transport.ReversibleConnection.Flat;Komorebi.PaperIII.Transport.ReversibleConnection.canonicalConnection;Komorebi.PaperIII.Transport.ReversibleConnection.canonicalConnection_flat`

The words *calibrated* and *flat* have only the algebraic meanings displayed above.

Remark 5.7 (Graph-local variant). One may specify Γ_e only on the oriented edges of a chart graph, together with inverse maps on reversed edges. Path transport and loop holonomy below still make sense. A direct cocycle defect for (a, b, c), however, requires a declared direct transport from a to c. The pairwise API in definition 5.5 keeps this comparison total.

5.3 Cocycle defect and triangle holonomy

The commutative-ring hypothesis makes subtraction of linear maps available from this point through section 7.

Definition 5.8 (Cocycle defect)**.** For $a, b, c \in B$, define the additive cocycle defect

$$\Delta_{c,b,a} = \Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} - \Gamma_{c\leftarrow a} \in \mathrm{Hom}_R(C_a, C_c).$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.ReversibleConnection.cocycleDefect`

The manuscript subscript order is target, intermediate, source. The Lean arguments are supplied in traversal order `(a,b,c)`.

Definition 5.9 (Triangle holonomy)**.** The triangle holonomy based at a is

$$H_{c,b,a} = \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} \in \mathrm{Aut}_R(C_a).$$

It traverses $a \to b \to c \to a$.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.ReversibleConnection.triangleHolonomy;Komorebi.PaperIII.Transport.ReversibleConnection.triangleObstruction`

Holonomy is attached to the chosen connection Γ, never to an unspecified coordinate change.

Theorem 5.10 (Defect–holonomy identity)**.** *For every triple* (a, b, c)*,*

$$\boxed{H_{c,b,a} - \mathrm{id}_{C_a} = \Gamma_{a\leftarrow c}\Delta_{c,b,a}.}$$

Consequently,

$$\Delta_{c,b,a} = 0 \quad \Longleftrightarrow \quad H_{c,b,a} = \mathrm{id}_{C_a}.$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.ReversibleConnection.triangleHolonomy_sub_id;Komorebi.PaperIII.Transport.ReversibleConnection.cocycleDefect_eq_zero_iff_triangleHolonomy_eq_refl`

Proof. Using reversibility,

$$\begin{aligned}\Gamma_{a\leftarrow c}\Delta_{c,b,a} &= \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} - \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow a}\\ &= H_{c,b,a} - \mathrm{id}_{C_a}.\end{aligned}$$

Left composition by the equivalence $\Gamma_{a\leftarrow c}$ is injective, which gives the equivalence of the two vanishing statements. □

Corollary 5.11 (Canonical transport has zero connection defect)**.** *If $\Gamma_{b\leftarrow a} = J_{b\leftarrow a}$, then*

$$\Delta_{c,b,a} = 0, \qquad H_{c,b,a} = \mathrm{id}$$

for all triples.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.ReversibleConnection.canonicalTransport_zero_connectionDefect; Komorebi.PaperIII.Transport.ReversibleConnection.canonicalTransport_triangleHolonomy_eq_refl`

Proof. The defect vanishes by the canonical cocycle in theorem 4.4; apply theorem 5.10. □

5.4 Paths and endpoint independence

Definition 5.12 (Path-ordered connection transport)**.** For a vertex sequence $p = (a_0, a_1, \ldots, a_n)$, define

$$\Gamma_p = \Gamma_{a_n\leftarrow a_{n-1}} \cdots \Gamma_{a_1\leftarrow a_0}, \qquad \Gamma_{(a_0)} = \mathrm{id}_{C_{a_0}}.$$

If $a_n = a_0$, then Γ_p is the holonomy of the loop p.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.ReversibleConnection.pathTransport;Komorebi.PaperIII.Transport.ReversibleConnection.VertexPath;Komorebi.PaperIII.Transport.ReversibleConnection.indexedPathTransport`

Products are path-ordered: the first traversed edge acts on the right.

Proposition 5.13 (Flat pair transport collapses to endpoints)**.** *If Γ is flat, then every path satisfies*

$$\Gamma_p = \Gamma_{a_n\leftarrow a_0}.$$

In particular, every loop has identity holonomy.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.ReversibleConnection.pathTransport_eq_direct_of_flat;Komorebi.PaperIII.Transport.ReversibleConnection.indexedPathTransport_eq_direct_of_flat;Komorebi.PaperIII.Transport.ReversibleConnection.indexedLoopTransport_eq_refl_of_flat`

Proof. Induct on the path length. Appending a_{n+1} left-multiplies the inductive expression by $\Gamma_{a_{n+1}\leftarrow a_n}$, and flatness replaces the two factors by $\Gamma_{a_{n+1}\leftarrow a_0}$. For a loop, normalization gives $\Gamma_{a_0\leftarrow a_0} = \mathrm{id}$. □

Remark 5.14. The converse *trivial holonomy implies flatness* depends on which loops are present in a graph-local API. In the complete pair API, triviality of every triangle holonomy is equivalent to flatness by theorem 5.10. No topological simple-connectivity theorem is being invoked.

6 Gauge covariance and comparison with canonical transport

Definition 6.1 (Gauge change)**.** A gauge is a family of automorphisms $G_a \in \mathrm{Aut}_R(C_a)$. It acts on a connection by

$$\Gamma^G_{b\leftarrow a} = G_b \Gamma_{b\leftarrow a} G_a^{-1}.$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.ReversibleConnection.Gauge;Komorebi.PaperIII.Transport.ReversibleConnection.gauge`

A gauge changes coordinate representatives at each chart; it does not change the chart graph.

Theorem 6.2 (Gauge covariance)**.** *The defect and triangle holonomy transform as*

$$\Delta^G_{c,b,a} = G_c \Delta_{c,b,a} G_a^{-1}, \qquad H^G_{c,b,a} = G_a H_{c,b,a} G_a^{-1}.$$

Thus flatness and triviality of triangle holonomy are gauge invariant.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.ReversibleConnection.gauge_cocycleDefect;Komorebi.PaperIII.Transport.ReversibleConnection.gauge_triangleHolonomy;Komorebi.PaperIII.Transport.ReversibleConnection.gauge_flat_iff;Komorebi.PaperIII.Transport.ReversibleConnection.gauge_triangleHolonomy_eq_refl_iff`

Proof. Substitute the definition of Γ^G. Adjacent factors $G_b^{-1}G_b$ and $G_c^{-1}G_c$ cancel. In the defect, both the composite and direct terms acquire the same outer factors G_c and G_a^{-1}. In the loop, all inner factors cancel and only conjugation at the base remains. □

Theorem 6.3 (Flat connections are gauges of the canonical pair transport)**.** *Suppose the C_a are exact packet-coordinate modules for one ambient module, so the canonical $J_{b\leftarrow a}$ is available. If Γ is a flat normalized reversible pair connection, choose $o \in B$ and set*

$$G_a = \Gamma_{a\leftarrow o} J_{o\leftarrow a}.$$

Then every pair transport satisfies

$$\boxed{\Gamma_{b\leftarrow a} = G_b J_{b\leftarrow a} G_a^{-1}.}$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.ReversibleConnection.flatGauge;Komorebi.PaperIII.Transport.ReversibleConnection.flat_isGaugeOfCanonical`

Proof. Reversibility gives $G_a^{-1} = J_{a\leftarrow o}\Gamma_{o\leftarrow a}$. Therefore

$$\begin{aligned} G_b J_{b\leftarrow a} G_a^{-1} &= \Gamma_{b\leftarrow o} J_{o\leftarrow b} J_{b\leftarrow a} J_{a\leftarrow o} \Gamma_{o\leftarrow a} \\ &= \Gamma_{b\leftarrow o}\Gamma_{o\leftarrow a} \\ &= \Gamma_{b\leftarrow a}. \end{aligned}$$

The first reduction uses the canonical cocycle twice; the second uses flatness of Γ. □

Boundary 6.4 (Flat does not mean calibrated). The theorem produces a gauge relating Γ to J. It does not prove $\Gamma = J$ before that gauge is chosen. Conversely, nontrivial triangle holonomy cannot be removed by a gauge, because conjugation cannot turn a nonidentity automorphism into the identity.

7 Obstruction packets: local certificates and holonomy blocks

7.1 The generic compiled certificate

Let Candidate, Packet, and Residual be arbitrary types, with a distinguished $0 \in \mathsf{Residual}$, and let

$$r : \mathsf{Candidate} \longrightarrow \mathsf{Packet} \longrightarrow \mathsf{Residual}$$

be a residual function.

Definition 7.1 (Candidate-specific solution and obstruction packet)**.** A candidate x is a solution when

$$\mathrm{IsSolution}_r(x) \quad \Longleftrightarrow \quad \forall p,\ r(x,p) = 0.$$

An obstruction packet for x is a pair consisting of a named packet p and a certificate $r(x,p) \neq 0$.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.NonlinearObstructions.IsSolution;Komorebi.PaperIII.NonlinearObstructions.ObstructionPacket`

Theorem 7.2 (Exact logical force of a local obstruction)**.** *Every obstruction packet for x proves that x is not a solution, and*

$$\neg\, \mathrm{IsSolution}_r(x) \quad \Longleftrightarrow \quad \textit{there exists an obstruction packet for } x.$$

The reverse implication from failure to a packet witness is classical and does not require the packet type to be finite.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.NonlinearObstructions.ObstructionPacket.not_isSolution;Komorebi.PaperIII.NonlinearObstructions.obstructionPacket_of_not_isSolution;Komorebi.PaperIII.NonlinearObstructions.not_isSolution_iff_nonempty_obstructionPacket`

Proof. A purported solution would make the residual vanish at the certified packet, contradicting its nonzero field. Conversely, if no packet carries a nonzero residual, then every residual is zero and the candidate satisfies the definition of solution. □

Boundary 7.3 (Candidate-specific means candidate-specific)*.* The theorem rules out the named x. It neither constructs a solution nor shows that every candidate is obstructed. It provides no minimality, stability, computability, or classification theorem for the packet witness.

7.2 Holonomy-block specialization

For a reversible connection, the generic residual may be specialized at a fixed triangle by taking a packet label to be a pair (μ, ν) and the residual to be the corresponding block of $H - \mathrm{id}$. A block is therefore a packet residual; when it is nonzero, the label together with a proof of nonvanishing is an obstruction-packet certificate in the sense of definition 7.1. The reconstruction theorem below is the extra statement needed to turn vanishing of all those local residuals into the cocycle equation at that triangle.

Fix a triangle (a, b, c) and resolve the base fiber C_a by a finite complete family of idempotents $(\Pi^a_\mu)_{\mu \in I_a}$. Orthogonality may also hold, as it does for the exact range-coordinate projectors below, but the compiled reconstruction theorem does not require it. In exact range coordinates, Π^a_μ is simply the projector that retains component μ and zeros the other components.

Definition 7.4 (Holonomy obstruction and packet residuals)**.** Define

$$\mathrm{Ob}_{c,b,a} = H_{c,b,a} - \mathrm{id}_{C_a}, \qquad \mathrm{Ob}^{\mu\nu}_{c,b,a} = \Pi^a_\mu \,\mathrm{Ob}_{c,b,a}\, \Pi^a_\nu.$$

The map $\mathrm{Ob}^{\mu\nu}_{c,b,a}$ is the obstruction block, or packet residual, from input packet ν to output packet μ. If it is nonzero, then (μ, ν) and that nonvanishing proof form an obstruction-packet certificate. Diagonal blocks detect within-packet holonomy; off-diagonal blocks detect holonomy-induced packet mixing.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.HolonomyBlocks.PacketResolution;Komorebi.PaperIII.HolonomyBlocks.PacketResolution.block`

An obstruction-packet certificate here is a nonzero block of a precisely specified holonomy defect together with its packet-pair label; it is not merely a heuristic label for a large residual.

Theorem 7.5 (Exact obstruction-block reconstruction and detection)**.** *The total obstruction has the exact finite resolution*

$$\boxed{\mathrm{Ob}_{c,b,a} = \sum_{\mu,\nu \in I_a} \mathrm{Ob}^{\mu\nu}_{c,b,a}\,.}$$

Moreover, the following are equivalent:

(i) $\Delta_{c,b,a} = 0$*;*

(ii) $H_{c,b,a} = \mathrm{id}$*;*

(iii) $\mathrm{Ob}_{c,b,a} = 0$*;*

(iv) $\mathrm{Ob}^{\mu\nu}_{c,b,a} = 0$ *for every* μ, ν*.*

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.HolonomyBlocks.obstruction_sum_blocks;Komorebi.PaperIII.HolonomyBlocks.obstructionBlocks_zero_iff;Komorebi.PaperIII.HolonomyBlocks.obstructionDetection_equivalences`

Proof. Insert completeness on both sides:

$$\mathrm{Ob} = \left(\sum_\mu \Pi^a_\mu\right) \mathrm{Ob} \left(\sum_\nu \Pi^a_\nu\right) = \sum_{\mu,\nu} \Pi^a_\mu \,\mathrm{Ob}\, \Pi^a_\nu.$$

Thus (iv) implies (iii), and (iii) implies (iv) by composition with the projectors. Conditions (ii) and (iii) are equivalent by the definition of Ob and additive cancellation. Conditions (i) and (ii) are equivalent by theorem 5.10. □

Figure 3 depicts the two-sided finite block resolution behind the reconstruction identity and its off-diagonal witnesses.

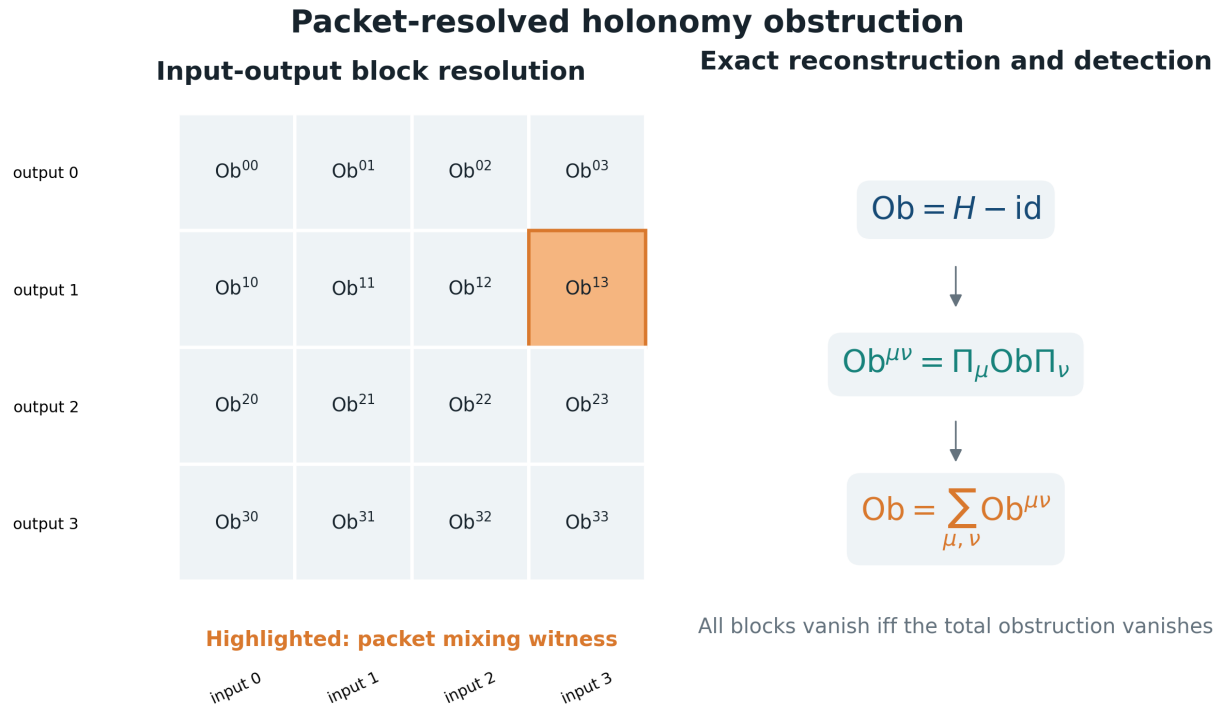


Figure 3: A finite complete packet resolution reconstructs the total holonomy obstruction from its input-output blocks. A nonzero off-diagonal block is an exact packet-mixing witness.

Corollary 7.6 (Packet-supported witness)**.** *If* $\mathrm{Ob}^{\mu\nu}_{c,b,a} \neq 0$*, then there is a vector* x *supported in packet* ν *such that*

$$\Pi^a_\mu \,\mathrm{Ob}_{c,b,a}\, x \neq 0.$$

Such an x *is an exact finite witness of nonflatness.*

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.HolonomyBlocks.exists_packetWitness_of_obstructionBlock_ne_zero`

Proof. Classically choose y on which the nonzero linear map $\Pi^a_\mu \,\mathrm{Ob}\, \Pi^a_\nu$ is nonzero and set $x = \Pi^a_\nu y$. Idempotence gives $\Pi^a_\nu x = x$, and the required output is the chosen nonzero value. □

7.3 Gauge behavior of packet blocks

Vanishing or nonvanishing of the total triangle obstruction is gauge invariant by theorem 6.2, but a fixed packet pattern need not be. A general G_a can mix the projectors Π^a_μ.

Proposition 7.7 (Packet-preserving gauge covariance)**.** *If* $G_a \Pi^a_\mu = \Pi^a_\mu G_a$ *for every* μ*, then*

$$\left(\mathrm{Ob}^G\right)^{\mu\nu} = G_a \,\mathrm{Ob}^{\mu\nu}\, G_a^{-1}.$$

Consequently the zero/nonzero status of each obstruction block is preserved by packet-preserving gauges.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.HolonomyBlocks.obstructionBlock_gauge_of_commutes;Komorebi.PaperIII.Holonomy Blocks.obstructionBlock_gauge_eq_zero_iff`

Proof. Use $\mathrm{Ob}^G = G_a \,\mathrm{Ob}\, G_a^{-1}$ and commute G_a and G_a^{-1} through the two packet projectors. Composition by an equivalence is injective in both directions. □

Boundary 7.8 (Packet patterns are not absolute). Without the commutation hypothesis in proposition 7.7, only the vanishing of the total obstruction is gauge invariant. The location of individual nonzero blocks may change under a gauge that mixes packets.

8 Packet-supported products and holonomy walks

The word-to-walk mechanism is not special to powers of one matrix. Let V be one finite packet type, let R be a semiring, and let $B_1, \dots, B_m \in R^{V\times V}$ be a finite sequence of possibly different packet matrices. Rows are targets and columns are sources as before.

Proposition 8.1 (Heterogeneous packet-path expansion). *For $i, j \in V$ and $m \geq 1$, set $i_0 = i$ and $i_m = j$. Then*

$$\boxed{(B_m \cdots B_1)_{ji} = \sum_{(i_1,\dots,i_{m-1})\in V^{m-1}} B_m[i_m, i_{m-1}] \cdots B_1[i_1, i_0].}$$

When $m = 1$, the sum is over the unique empty tuple and its summand is $B_1[j, i]$. For $m = 0$, the product is the identity and the entry is the Kronecker delta δ_{ji}. If every $B_r[q, p]$ vanishes outside a declared relation $\mathcal{R}_r(p, q)$, then the endpoint entry is zero whenever there is no packet path compatible with the successive relations.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.EdgeOperators.heterogeneousProduct;Komorebi.PaperIII.EdgeOperators.heterogeneousPathSum;Komorebi.PaperIII.EdgeOperators.heterogeneousPacketPathExpansion;Komorebi.PaperIII.EdgeOperators.SupportedStep;Komorebi.PaperIII.EdgeOperators.SupportedPath;Komorebi.PaperIII.EdgeOperators.heterogeneousProduct_apply_eq_zero_of_no_supportedPath`

Proof. Expand each matrix product entry and distribute the finite sums. Induction on the factor list yields the displayed path sum, with the empty list evaluated as the identity matrix. If no supported intermediate packet exists, every summand contains an entry that is zero by its declared support relation. □

Boundary 8.2 (Changing packet types along a chain). The compiled theorems `heterogeneousPacketPathExpansion_two` and `heterogeneousPacketPathExpansion_three` allow different packet types at the successive stages of two- and three-factor products. The arbitrary-length theorem above permits different factors but uses one common finite packet type V. A fully dependent arbitrary-length chain of changing modules and packet types remains outside the version 1.0.1 contract.

Remark 8.3 (Checked specialization). When every B_r is one repeated endomorphism T, the recursive packet-power specialization is inherited from Paper I.

Formal status: inherited Lean-backed result.
`Komorebi.PacketNaturalityCompletion.Packets.transferBlock_pow_eq_packetPathSum`

For a loop represented on one common finite scalar packet-coordinate type, the proposition resolves its successive transport matrices into packet walks. If R is a ring, as in the reversible-connection API, then $(H - I)_{ji} = H_{ji} - \delta_{ji}$; in particular, off-diagonal entries are unchanged. Thus a nonzero off-diagonal holonomy obstruction in this common-index model requires a transport word along the closed chart loop whose packet path runs from its input packet to its output packet. This conclusion is conditional on declared block supports for the connection matrices; an arbitrary connection need not respect the transfer graph from section 2.

9 A minimal exact example

Example 9.1 (A scalar triangle)**.** Let $B = \{0, 1, 2\}$ and $C_0 = C_1 = C_2 = \Bbbk$ for a field $\Bbbk$. Choose $u, v, w \in \Bbbk^\times$ and set

$$\Gamma_{1\leftarrow 0}(x) = ux, \qquad \Gamma_{2\leftarrow 1}(x) = vx, \qquad \Gamma_{2\leftarrow 0}(x) = wx,$$

with reverse transports given by inverse scalars. Then

$$\Delta_{2,1,0}(x) = (vu - w)x,$$

and the triangle holonomy based at 0 is

$$H_{2,1,0}(x) = w^{-1}vux.$$

Therefore

$$H_{2,1,0} - \mathrm{id} = w^{-1}\Delta_{2,1,0}.$$

The connection is flat on this triangle exactly when $vu = w$. If the three coordinate systems are literally the same and canonical transport is used, then $u = v = w = 1$ and the obstruction vanishes. Choosing arbitrary nonzero u, v, w is a connection choice, not a consequence of changing exact packet coordinates.

Example 9.2 (A two-step transfer walk)**.** Let a packet graph have edges $e : 0 \to 1$ and $f : 1 \to 2$, with no other edges. Then

$$T = L_e + L_f, \qquad (T^2)_{20} = c(f)c(e).$$

The products L_eL_e, L_eL_f, and L_fL_f vanish by typed mismatch; only L_fL_e, corresponding to the walk $0 \to 1 \to 2$, contributes to the $(2, 0)$ entry. This example also fixes the noncommutative coefficient order.

10 Canonical coordinates for arbitrary maps

The analysis and synthesis identities are identities of functions, not only of linear maps. Consequently exact coordinate change does not become curved merely because the represented operation is nonlinear.

Proposition 10.1 (Set-level exact naturality)**.** *Let $\mathcal{A}^X_a : X \xrightarrow{\sim} X_a$ and $\mathcal{A}^X_b : X \xrightarrow{\sim} X_b$ be exact coordinate equivalences for X, with inverses $\mathcal{S}^X_a$ and $\mathcal{S}^X_b$. Likewise, let $\mathcal{A}^Y_r : Y \xrightarrow{\sim} Y_r$ and $\mathcal{A}^Y_s : Y \xrightarrow{\sim} Y_s$ be exact coordinate equivalences for Y, with inverses $\mathcal{S}^Y_r$ and $\mathcal{S}^Y_s$. Let $F : X \to Y$ be any function, with no linearity or differentiability hypothesis. Set*

$$J^X_{b\leftarrow a} = \mathcal{A}^X_b \circ \mathcal{S}^X_a, \qquad J^Y_{s\leftarrow r} = \mathcal{A}^Y_s \circ \mathcal{S}^Y_r,$$

and define

$$\widehat{F}_{r\leftarrow a} = \mathcal{A}^Y_r \circ F \circ \mathcal{S}^X_a.$$

Then

$$J^Y_{s\leftarrow r} \circ \widehat{F}_{r\leftarrow a} = \widehat{F}_{s\leftarrow b} \circ J^X_{b\leftarrow a}.$$

If $\mathcal{A}^Z_t : Z \xrightarrow{\sim} Z_t$ is another exact coordinate equivalence and $G : Y \to Z$ is any function, then exact coordinate representation also preserves composition:

$$\widehat{G \circ F}_{t\leftarrow a} = \widehat{G}_{t\leftarrow r} \circ \widehat{F}_{r\leftarrow a}.$$

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.Transport.exactCoordinateMap;Komorebi.PaperIII.Transport.exactCoordinateMap_comp;Komorebi.PaperIII.Transport.exactCoordinateMap_naturality`

Proof. Both sides of the naturality equation reduce to $\mathcal{A}_s^Y \circ F \circ \mathcal{S}_a^X$ using $\mathcal{S} \circ \mathcal{A} = \mathrm{id}$ in X and Y. For composition, the middle $\mathcal{S}^Y \circ \mathcal{A}^Y$ cancels in the same way. □

Boundary 10.2 (Where nonlinear defects can enter). A nonlinear *coordinate* defect cannot be blamed on the exact canonical J of definition 4.3; proposition 10.1 makes that defect identically zero. Nonzero defects require extra structure such as a noncanonical connection, truncation, approximation, restricted domains, or failure of an operation to respect a declared packet model.

11 Nonlinear operation trees: proved locality and scoped roadmap

11.1 The compiled unary–binary foundation

Definition 11.1 (Finite operation-tree grammar)**.** For types of input labels I, constants K, unary symbols U, and binary symbols B, the released formal grammar has four constructors:

$$\begin{aligned} t ::= {} & \mathrm{input}(i) \mid \mathrm{constant}(k) \mid \mathrm{unary}(u, t) \\ & \mid \mathrm{binary}(b, t_1, t_2). \end{aligned}$$

Given an arbitrary value type Y, evaluation takes completely explicit maps

$$I \to Y, \qquad K \to Y, \qquad U \to Y \to Y, \qquad B \to Y \to Y \to Y.$$

No algebraic, continuity, or differentiability law is imposed on these semantics.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.NonlinearObstructions.OperationTree;Komorebi.PaperIII.NonlinearObstructions.OperationTree.eval`

When I has decidable equality, let $\mathrm{inputs}(t)$ be the finite set of input labels actually occurring in t.

Theorem 11.2 (Operation-tree input locality)**.** *Let $\eta_1, \eta_2 : I \to Y$ be two input environments. If*

$$\eta_1(i) = \eta_2(i) \qquad \text{for every } i \in \mathrm{inputs}(t),$$

then the two evaluations of t are equal under the same constant, unary, and binary semantics. In particular, if $\mathrm{inputs}(t) = \varnothing$, evaluation is independent of the input environment.

Formal status: Lean-backed in Paper III.
`Komorebi.PaperIII.NonlinearObstructions.OperationTree.eval_congr_on_inputs;Komorebi.PaperIII.NonlinearObstructions.OperationTree.eval_eq_of_inputs_eq_empty`

Proof. Structural induction on t. The input case is the agreement hypothesis; the constant case is reflexive; the unary case applies the induction hypothesis to its child; and the binary case applies it independently to the left and right subtrees, whose input sets form a union. □

Boundary 11.3 (Exact force of locality). The theorem says evaluation reads only syntactically named inputs. It does not prove that the operations are linear or nonlinear in any analytic sense, that the tree is stable, that two different trees are equivalent, or that a solution to a tree equation exists.

11.2 Typed colored extension (roadmap)

Definition 11.4 (Finitary colored operation signature)**.** A finitary colored signature consists of:

- a finite type V of colors;
- a type Σ of operation symbols;
- an arity $r(\sigma) \in \mathbb{N}$ for every $\sigma \in \Sigma$;
- input colors $\mathrm{in}_\sigma : \mathrm{Fin}(r(\sigma)) \to V$;
- an output color $\mathrm{out}(\sigma) \in V$.

A realization assigns a space X_v to each color and a map

$$\Phi_\sigma : \prod_{q \in \mathrm{Fin}(r(\sigma))} X_{\mathrm{in}_\sigma(q)} \longrightarrow X_{\mathrm{out}(\sigma)}.$$

Formal status: roadmap only. The intended Lean target is a dependent inductive typed tree; no such declaration is part of the version 1.0.1 completion contract.

Definition 11.5 (Typed operation tree and evaluation)**.** A typed operation tree is recursively either a leaf of declared color or a node labeled by σ, with one correctly colored child for each input slot of σ. Evaluation is recursive:

$$\mathrm{ev}(\mathrm{node}(\sigma; t_1, \ldots, t_{r(\sigma)})) = \Phi_\sigma(\mathrm{ev}(t_1), \ldots, \mathrm{ev}(t_{r(\sigma)})).$$

Formal status: roadmap only. Candidate API: `Komorebi.PaperIII.NonlinearObstructions.ColoredOperationTree` and `ColoredOperationTree.eval`. This is stronger than the compiled uncolored unary–binary grammar.

11.3 Packet labelings and the multilinear first target

Suppose every X_v has a finite complete packet family $(P_i^v)_{i \in I_v}$. A packet labeling assigns an admissible packet index to each edge or node of a typed tree. For multilinear Φ_σ, inserting

$$\mathrm{id}_{X_v} = \sum_{i \in I_v} P_i^v$$

at every input and output yields a finite sum over packet-labeled trees. This is the branching analogue of inserting intermediate projectors in proposition 8.1.

Roadmap item 11.6 (Finite multilinear tree expansion)*.* The first milestone for the nonlinear module is an exact theorem saying that the evaluation of a finite tree of multilinear nodes equals the finite sum of its packet-resolved labeled evaluations. Required proof obligations include:

(a) a dependent well-typed tree syntax;

(b) finite packet-label types at every color;

(c) an explicit convention for nullary and unary nodes;

(d) multilinearity in every input slot;

(e) preservation of input-slot order without assuming commutativity of operations.

Formal status: roadmap only. Candidate target: `Komorebi.PaperIII.NonlinearObstructions.ColoredOperationTree.eval_eq_sum_packetLabellings`.

11.4 Connection defects at nonlinear nodes

For coordinate models of the source and target colors, an additive target coordinate space, and a chosen noncanonical connection, a pointwise node defect can be defined by

$$\mathfrak{D}^{\Gamma}_{b,a}(\Phi) = \Gamma^{Y}_{b\leftarrow a} \circ \widehat{\Phi}_a - \widehat{\Phi}_b \circ \left(\prod_q \Gamma^{X_q}_{b\leftarrow a} \right).$$

This expression compares two *chosen* routes. With canonical exact transport in every slot it vanishes by proposition 10.1.

Roadmap item 11.7 (Operation-tree defect propagation). For multilinear nodes, one expects an exact finite expansion obtained by replacing one or more nodes by their defects and transporting the remaining branches. For smooth nonlinear nodes, the corresponding chain rule is indexed by rooted partitions and Faà di Bruno trees. A formal theorem will require a fixed differentiability category, base points, derivative conventions, symmetry factors, and remainder policy.

Formal status: roadmap only. No tree-level defect identity, higher-chain-rule theorem, or nonlinear norm bound is claimed here.

Boundary 11.8 (Analytic and geometric exclusions). This version does not prove convergence of an infinite tree series, existence of derivatives, a Banach-space Faà di Bruno theorem, a Baker–Campbell–Hausdorff expansion, a continuum limit, or an identification of algebraic holonomy with curvature of a smooth bundle. It also does not derive a connection from data, optimization, physics, or probability.

12 Formalization ledger for version 1.0.1

Topic	Status	Declaration or roadmap target	Exact boundary
Packet matrix and edge operator	Lean-backed III	`PacketGraph.packetMatrix`; `EdgeSystem.edgeOp`	Finite scalar matrices over a semiring; one scalar coordinate per packet.
Edge support and pair dichotomy	Lean-backed III	`edgeOp_eq_zero_off_support`; `edgeOp_mul_edgeOp_eq_zero`; `edgeOp_mul_edgeOp_of_composable`	Broken interfaces vanish; matched pairs collapse to the endpoint block with coefficient in operator order.
Typed word evaluation	Lean-backed III	`Composable`; `wordOp`; `wordOp_append`; `wordOp_eq_zero_of_not_composable`	List order is execution order; concatenation reverses operator-product order; nonwalks vanish.
Finite word and walk expansions	Lean-backed III	`explicitWordExpansion`; `walkExpansion`; `wordExpansion_eq_explicitWordExpansion`; `transferOp_pow_eq_walkExpansion`	Actual finite tuple sum and zero-filtered composable-walk sum; finite graph and packet types.

Topic	Status	Declaration or roadmap target	Exact boundary
Endpoint coefficient walk sum	Lean-backed III	`reverseCoefficient`; `Connects`; `transferOp_pow_apply_eq_sum_walks`	Exact scalar endpoint entry, including the empty-walk case and noncommutative coefficient order.
Fixed reachability and orbit support	Lean-backed III; inherited hull bridge	`transferOp_pow_apply_eq_zero_of_no_walk`; `transferOp_pow_mulVec_apply_eq_zero_of_unreachable_support`	Finite powers and supported seeds; the orbit-hull span conclusion uses the frozen Paper I contract.
Exact frame transport	Lean-backed III	`exactTransport_cocycle`; `exactPathTransport_eq_endpoint`; `exactPathTransport_loop`	One frame family on a single module; arbitrary finite paths telescope and closed paths are trivial.
Exact coordinate cocycle	Inherited Lean-backed II	`NestedRefinement.refineCoordinates_comp`; `NestedRefinement.coordinateRefinementEquiv_trans`	Complete finite projector coordinates, not reduction.
Typed coordinate naturality	Inherited Lean-backed II	`NestedRefinement.coordinateOperator_comp`; `NestedRefinement.coordinateOperator_naturality`	Linear maps over semiring modules.
Minimal connection API	Lean-backed III	`Transport.Connection`; `cocycleDefect`; `triangleHolonomy`; `triangleHolonomy_eq_refl_of_flat`	Only self-normalization is assumed; reverse inversion and the cocycle are not structure fields.
Twisted three-point loop	Lean-backed III	`twistedThreePointConnection`; `twisted_triangleHolonomy_ne_refl`	Concrete rational connection; the selected loop sends 1 to 2.
Reversible additive defect	Lean-backed III	`ReversibleConnection`; `triangleHolonomy_sub_id`; `cocycleDefect_eq_zero_iff_triangleHolonomy_eq_refl`	Heterogeneous module fibers, reverse-edge inversion, and an additive defect over a commutative ring.
Reversible finite paths	Lean-backed III	`pathTransport_eq_direct_of_flat`; `indexedPathTransport_eq_direct_of_flat`; `indexedLoopTransport_eq_refl_of_flat`	Finite pair-connection paths collapse to endpoints under algebraic flatness.
Gauge covariance	Lean-backed III	`gauge_cocycleDefect`; `gauge_triangleHolonomy`; `gauge_flat_iff`	Fiberwise algebraic conjugation; no smooth gauge theory.

Topic	Status	Declaration or roadmap target	Exact boundary
Flat versus canonical transport	Lean-backed III	`flatGauge`; `flat_isGaugeOfCanonical`	A chosen base chart and ambient frames exhibit every flat pair connection as a gauge transform of canonical transport.
Operation-tree locality	Lean-backed III	`OperationTree`; `OperationTree.eval`; `OperationTree.eval_congr_on_inputs`	Arbitrary unary and binary semantics; only dependence on named inputs is proved.
Generic obstruction packet	Lean-backed III	`ObstructionPacket.not_isSolution`; `not_isSolution_iff_nonempty_obstructionPacket`	Exact candidate-specific logical certificate; no global classification.
Holonomy block obstruction	Lean-backed III	`obstruction_sum_blocks`; `obstructionBlocks_zero_iff`; `obstructionDetection_equivalences`	Finite complete idempotent packet family on the base fiber; orthogonality is not required by the compiled reconstruction.
Packet witness and block gauge	Lean-backed III	`exists_packetWitness_of_obstructionBlock_ne_zero`; `obstructionBlock_gauge_of_commutes`; `obstructionBlock_gauge_eq_zero_iff`	Witness extraction is classical; blockwise covariance requires the gauge to commute with every packet projector.
Heterogeneous matrix paths	Lean-backed III	`heterogeneousPacketPathExpansion`; `heterogeneousProduct_apply_eq_zero_of_no_supportedPath`	Arbitrarily many different matrices over one common finite packet type; fully typed changing-index chains compile only at lengths two and three.
Arbitrary-function coordinates	Lean-backed III	`exactCoordinateMap_comp`; `exactCoordinateMap_naturality`	Exact set-level equivalences and arbitrary functions; no linearity or differentiability assumption.
Colored and smooth trees	Roadmap only	roadmap `ColoredOperationTree.eval_eq_sum_packetLabellings`	Dependent colors, multilinear packet expansion, derivatives, and Faà di Bruno identities remain outside the contract.

12.1 Kernel-backed Paper III surface

The authoritative declaration-level contract is `formalization/lean/LEAN_THEOREM_MAP_v1.0.1.md`. It groups the compiled surface as follows.

- **EdgeOperators and EdgeExtensions:** scalar packet matrices, typed edge and word support, finite word/walk expansions, endpoint coefficient sums, fixed-length reachability, orbit support, and common-index heterogeneous products with declared supports.
- **Transport and ReversibleConnections:** exact frame transport, the minimal connection, heterogeneous reversible pair transport, additive defect and triangle holonomy, finite paths, canonical connections, and the fiberwise gauge action.
- **CoordinateNaturality and HolonomyBlocks:** composition and naturality for arbitrary functions between exact coordinate systems, finite two-sided block reconstruction, equivalent obstruction criteria, packet-supported witnesses, and packet-preserving block gauges.
- **NonlinearObstructions:** finite unary–binary operation trees, input locality, and candidate-specific obstruction certificates.

The release verification report records the checker-generated declaration and audit totals. The parity gate requires every fully qualified Paper III target printed beside a statement to occur in the theorem map, rejects stale or duplicate targets, and checks it against the compiled environment. `KernelAudit.lean` reports dependencies for the mapped and supporting theorems, and its fail-closed gate scans every compiled `Komorebi.PaperIII` declaration. The four inherited statement contracts remain identified separately by their Papers I–II names.

Boundary 12.1 (No scope promotion by prose). The displayed endpoint coefficient formula, reachability results, reversible connection theorems, additive defect–holonomy identity, gauge covariance, holonomy-block reconstruction, packet witnesses, common-index heterogeneous path expansion, and arbitrary-function coordinate naturality are all part of the version 1.0.1 kernel-backed surface. What remains outside is the dependent fiber-valued edge-system generalization, a fully dependent arbitrary-length chain whose module and packet type change at every stage, dependent colored operation trees, packet-labeled multilinear tree expansions, nonlinear tree-defect and Faà di Bruno identities, and the analytic and geometric extensions listed above.

13 Conclusion

Typed packet support turns the expansion of a transfer power into a walk calculus: incompatible words vanish for structural reasons, not because of numerical cancellation. Exact packet-coordinate transport is even more rigid. It is the canonical analysis–synthesis equivalence and therefore satisfies its cocycle law exactly; every finite canonical loop telescopes to the identity. Holonomy enters only after a separate connection is supplied. The kernel-backed minimal API proves that flatness kills both the relative defect and holonomy, and it exhibits a nontrivial rational loop. Under the kernel-backed reversible API, a nonzero additive triangle defect is equivalent to nonidentity holonomy, transforms covariantly under gauge, and is detected exactly by a finite family of packet blocks. Finite flat paths collapse to their endpoints, and every flat reversible pair connection is explicitly gauge-equivalent to canonical transport through a chosen base chart.

These facts define a clean boundary for the nonlinear program. At the audited boundary, exact coordinates preserve composition and satisfy naturality for arbitrary functions in Paper III, while the typed linear packet-coordinate counterpart remains inherited from Paper II. Nontrivial nonlinear

defects must therefore be attached to additional modeling choices or to operation-specific packet compatibility. Colored operation trees, multilinear packet labelings, and Faà di Bruno expansions remain explicit future milestones rather than completed results.

References

[1] Komorebi P., *Projector-Resolved Seed Calculus I: Typed Seeds, Operator Grammars, and Exact Reconstruction*, version 1.2.1, 2026.

[2] Komorebi P., *Projector-Resolved Seed Calculus II: Polyphase Refinement, Weighted Transfer, and Dyadic–Odd Resolution*, version 1.2.1, 2026.

[3] Leonardo de Moura and Sebastian Ullrich, *The Lean 4 Theorem Prover and Programming Language*, in André Platzer and Geoff Sutcliffe (eds.), *Automated Deduction – CADE 28*, Lecture Notes in Computer Science, vol. 12699, pp. 625–635, Springer, Cham, 2021. DOI: 10.1007/978-3-030-79876-5_37.

[4] The mathlib Community, *The Lean mathematical library*, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pp. 367–381, Association for Computing Machinery, New York, NY, USA, 2020. DOI: 10.1145/3372885.3373824.